

Review on research achievements of biogas from anaerobic digestion

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ABSTRACT

With the rising demand for renewable energy and environmental protection, anaerobic digestion of biogas technology has attracted considerable attention within the scientific community. This paper presents a comprehensive review of research achievements on anaerobic digestion developments for biogas production. The review includes a discussion of factors affecting efficiency (temperature, pH, C/N ratio, OLR and retention time), accelerants (greenery biomass, biological pure culture and inorganic additives), reactors (conventional anaerobic reactors, sludge retention reactors and anaerobic membrane reactors) and biogas AD processes (lignocellulose waste, municipal solid waste, food waste, livestock manure and waste activated sludge) based on substrate characteristics and discusses the application of each forementioned aspect. The factors affecting efficiency are crucial to anaerobic digestion, because they play a major role in biogas production and determine the metabolic conditions for microorganism growth. As an additive, an accelerant is not only regarded as a nutrient resource, but can also improve biodegradability. The focus of reactor design is the sufficient utilization of a substrate by changing the feeding method and enhancing the attachment to biomass. The optimal digestion process balances the optimal digest conditions with the cost-optimal input/output ratio. Additionally, establishment of theoretical and technological studies should emphasize practicality based on laboratory-scale experiments because further development of biogas plants would allow for a transition from household to medium- and large-scale projects; therefore, improving stability and efficiency are recommended for advancing AD research.

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Abbreviations: AD, anaerobic digestion; GHG, greenhouse gas; VFA, volatile fatty acid; TAN, total ammonia nitrogen; AN, free ammonia; OLR, organic loading rate; SRT, solid retention time; HRT, hydraulic retention time; WAS, waste activated sludge; FW, food waste; OMSR, olive mill solid residue; COD, chemical oxygen demand; ASBR, anaerobic sequencing batch reactor; CSTR, anaerobic sequencing batch reactor; MBR, membrane bioreactor; APFR, anaerobic plug-flow reactor; ACR, anaerobic contact reactor; UASB, up-flow anaerobic sludge bed reactor; UASS, up-flow anaerobic solid-state reactor; ABR, anaerobic baffled reactor; CRT, cell retention time; IC, internal circulation reactor; BOD, biological oxygen demand; ITS, inclined tube settlers; FFB, fixed bed fixed film; AFBR, anaerobic fluidized bed reactor; AnMBR, anaerobic membrane bioreactor; AFBMR, anaerobic fluidized bed membrane reactor; GAC, granular activated carbon; PAC, powder activated carbon; EGSB, expanded granular sludge blanket; MSW, municipal solid waste; FW, food waste; WAS, waste activated sludge

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1. Introduction

The consumption of renewable energy is dramatically increasing, along with energy security concerns, efforts to mitigate the environmental impact of conventional fuels, and improvements in living standards and renewable technologies. Bioenergy can play a central role in promoting renewable alternatives. In fact, bioenergy is estimated to be the fourth largest energy resource in the world [1], and is a nearly GHG-neutral replacement for fossil fuels [2] due to its renewable and widely applicable characteristics and its abundance. Forestry resources, agricultural resources, sewage and industrial organic wastewater, municipal solid wastes, livestock and poultry dung and biogas are major categories for use.

Biogas, which is generally referring to gas from anaerobic digestion units, is a promising means of addressing global energy needs and providing multiple environmental benefits, as shown in Table 1 [3–7]. Examples include EU policy estimates that at least 25% of all bioenergy can be derived from biogas [8]; in Italy, 3405 GW h of electricity was produced from biogas in 2011 [9]; in Germany, approximately 4000 agricultural biogas production units were operated on German farms at the end of 2008, which is beneficial for farmer living-environment [10]; in China, 26.5 million biogas plants were built by 2007 with an output of 10.5 billion m³, and it was increased to 248 billion m³ (annually) by 2010 [11]. Furthermore, from a socio-economic point of view, biogas not only significantly reduces the costs of treating waste [8] but also has a relatively low feedstock cost. In addition, biogas has a lower sale price compared with diesel and petrol. These examples illustrate that biogas is utilized widely as a renewable source.

Biogas is generated from a digestion process under anaerobic conditions whose application is rapidly emerging as a viable means for providing continuous power generation. The AD cycle represents an integrated system of a physiological process of microbial and energy metabolism, as well as raw materials processing under

specific conditions (Fig. 1) [12]. However, the microbial community is sensitive to variations in the operating conditions applied. Thus, the AD process, if improperly managed, would become unstable and result in reduced biogas production. Although previous studies have discussed AD development, most focused on only one aspect (such as technology, mechanism, factors affecting efficiency, etc.) to minimize this instability (Table 2) [13–15] or on one substrate (such as livestock manure, urban solid waste, food waste, crop straw, etc.). An overall

Table 1
Biogas environmental benefits analysis.

Biogas	Corresponding contents	References
Green energy production	Electricity Heat Vehicle fuel Tri-generation	[5]
Organic waste disposal	Agricultural residues Industrial wastes Municipal solid wastes Household wastes Organic waste mixtures	
Environmental protection	Pathogen reduction through sanitation Less nuisance from insect flies Air & water pollution reduction Eutrophication and acidification reduction Forest vegetation conservation Replacing inorganic fertilizer	[3] [4]
Biogas-linked agrosystem	Livestock-biogas-fruit system Pig-biogas-vegetable greenhouse system Biogas-livestock and poultry farms system	[6] [7]
GHG emission reduction	Substituting conventional energy sources	[3]

The growth rate of microorganisms is significantly affected by pH changing. The relative abundance of microbial species has been observed to increase from 6 at pH 4.0 to 14 at pH 7.0 [20]. At pH 6.0, the dominant bacterial population is *Clostridium butyricum*, whereas at pH 8, the *Propionibacterium* species appears to prevail during anaerobic acidogenesis with a chemostat culture [21]. To reduce ammonia toxicity due to an increased concentration of free ammonia (FA), controlling the pH level to attain optimum microorganism growth represents one possible method. In a previous study, high extents of TSS and VSS degradation were obtained at pH levels of 7 and 8: 75% degradation of TSS and 85% degradation of VSS, whereas VFA showed no significant differences [22]. The VFA composition is also significantly affected by pH, as shown in Fig. 2 [20]. In another study, when pH level was set to 6.0, the hydrolytic enzyme activity was the highest, resulting in the highest VFA concentration, SCOD concentration and VFA/SCOD ratio and the lowest VS levels [23]. A significant positive correlation ($P=0.01$) has also been observed between hydrolysis and pH [24]. Therefore, the hydrolysis rate constant is considered to be pH-dependent. It should be emphasized that both methanogenic and acidogenic microorganisms have optimal pH levels. Methanogenesis is most efficient at pH 6.5–8.2, and the optimal pH is 7.0 [25]. The growth rate of methanogens is greatly reduced at pH levels below 6.6, and the activity of methanogenic bacteria decreased at a higher or lower pH [24]. The optimum pH of acidogenesis was between pH 5.5 and 6.5 [26] which is why a two-stage AD process separating the hydrolysis/acidification and acetogenesis/methanogenesis processes is the preferred mode of operation.

2.3. C/N ratio

The C/N ratio reflects the nutrient levels of a digestion substrate, and thus, digestion systems are sensitive to C/N ratio. A high C/N ratio induces a low protein solubilization rate and leads to low TAN and FA concentrations within a system. Thus ammonia inhibition may be avoided by optimizing the C/N ratio in the AD process. However, an excessively high C/N ratio provides insufficient nitrogen to maintain cell biomass and leads to fast nitrogen degradation by microbials, resulting in lower biogas production and vice versa. Substrates with an excessively low C/N ratio increase the risk of ammonia inhibition, which is toxic to methanogens and causes insufficient utilization of carbon sources. The optimal C/N ratio for anaerobic digestion has been shown to be between 20 and 30 or between 20 and 35, with a ratio of 25 being

the most commonly used [27–29]. Insufficient amounts of carbon or nitrogen can limit AD performance in the anaerobic mono-digestion of livestock manure or crop straw. Carbohydrates have been found to alleviate problems associated with insufficient substrate resources. The addition of carbohydrate matter appreciably enhances protein conversion and the protease activity of WAS [29]. Studies have shown that adjusting C/N ratio for swine manure digestion allows for maximum methane production with the addition either urea or glucose [30,31]. Although the application of this method would be rather easy, questions have been raised regarding the economic sustainability of this method in accelerating methane generation from large-scale digesters [30]. Co-digestion of agricultural waste with manure waste provides positive synergistic effects and can potentially dilute toxic compounds. C/N ratios of 15 and 20 at 35 °C and 55 °C, respectively, lead to significant ammonia inhibition. An approximately linear relationship was observed for co-digested wheat straw with swine manure as the C/N ratio increased from 16/1 to 25/1 ($R^2=0.9988$). The highest biogas yield of 341 mL/(g of VS added) was obtained from the co-digestion of swine manure and corn straw at a C/N ratio of 25. Similarly, C/N ratios of 25:1 and 30:1 provided the highest cumulative biogas production levels, approximately three-fold compared with a C/N ratio of 15:1 [32,33].

2.4. OLR

OLR represents the amount of volatile solids fed into a digester per day under continuous feeding. With increasing OLR, the biogas yield increases to an extent, but the equilibrium and productivity of the digestion process can also be greatly disturbed. Adding a large volume of new material daily may result in changes in the digester's environment and temporarily inhibits bacterial activity during the early stages of fermentation. This bacterial inhibition occurs due to an extremely high OLR leading to higher hydrolysis/acidogenesis bacterial activity than methanogenesis bacterial activity and thus increases VFA production, which eventually leads to an irreversible acidification. Thereafter, the pH of the digester decreases, and the hydrolysis process is inhibited such that the restricted methanogenesis bacteria are not able to convert as much VFA to methane. Hence, the maximum endurable OLR has been addressed and predicted in previous studies. For example, the optimal OLRs of 5 g VS/L/d, 9.2 kg VS/m³/d, 10.5 kg VS/m³/d and 9.2 g COD/L/d of WAS and FW co-digestion, sludge, FW and OMSR digestion and 5.2 g VS/L/d for relieving foam formation in

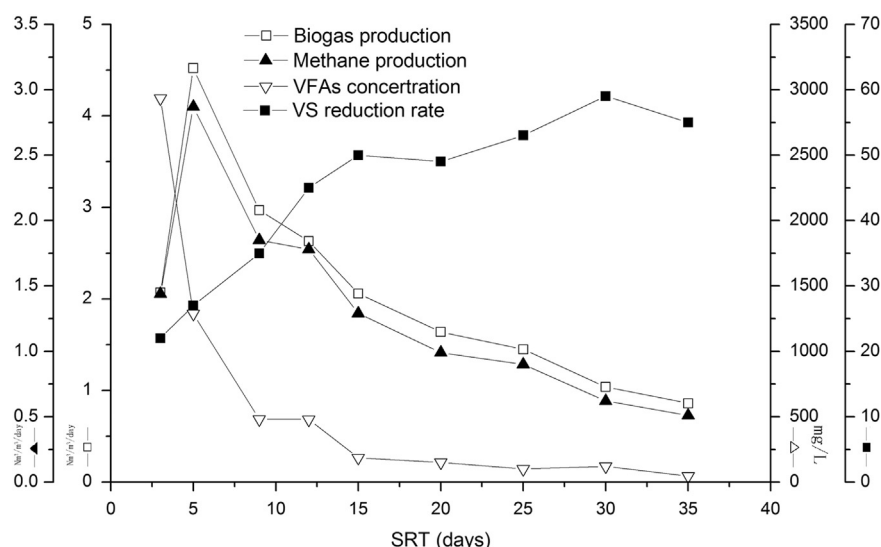


Fig. 3. Effect of SRT on AD progress stability and performance [42].

manure digester were observed under mesophilic conditions [34–37]. It should also be noted that the thermophilic system and effluent recirculation have great potential to relieve the overloading inhibition. Moreover, bacterial communities vary with OLR. The predominant bacteria are Firmicutes at low OLR, whereas Gammaproteobacteria, Actinobacteria, Bacteroidetes and Deferribacteres have been observed at high OLR [35]. In a previous study, the amount of Archaea increased as OLR increased from 1 to 2 kg COD/m³/d dairy of wastewater digestion [38].

2.5. Retention time

The retention time is the time required to complete the degradation of organic matter. It is associated with the microbial growth rate and depends on the process temperature, OLR and substrate composition. Two significant types of retention time are herein discussed: SRT, which is defined as the average time that bacteria (solids) spend in a digester, and HRT which is defined by the following equation [39]:

$$\text{HRT} = \frac{V}{Q}$$

where V is the biological reactor volume and Q the influent flow rate in time.

An average retention time of 15–30 days is required to treat waste under mesophilic conditions. Obtaining an effective HRT depends on the substrate composition and OLR; typically, a couple of weeks are necessary. Decreasing the HRT usually leads to VFA accumulation, whereas, a longer than optimal HRT results in insufficient utilization of digester components. For algal biomass, an HRT below 10 days results in low methane productivity [40]. The digestion stability of FW decreased at an 8-day HRT [41]. In summary, a low OLR and a long HRT provide the best strategy for achieving constant and maximal methane yields. Variations in the SRT destabilize and degrade the performance of anaerobic systems (Fig. 3) [42]. The figure shows that correlations between the SRT and biogas production rate ($r = -0.802^{**}$, $P = 0.009$), SRT and methane production rate ($r = -0.834^{**}$, $P = 0.005$), SRT and VS reduction rate ($r = 0.904^{**}$, $P = 0.001$), and SRT and TVFAs ($r = -0.714^{*}$, $P = 0.031$) are all significant. An increase in the SRT from 10 to 20 days caused a 25% decrease in specific gas production during AWS digestion [43]. The biogas production obtained at a 12-day SRT was tripled relative to that observed for a 35-day SRT. Process imbalance occurred as a result of foaming, VFAs accumulation and increased alkalinity at a 9-day SRT during the digestion of dewatered-sewage sludge [42].

3. Biogas AD accelerants

Many attempts have been made to increase gas production during the biogas AD process, including introduction of accelerants, i.e., biological and/or chemical additives. The adsorption of a substrate on the surface of such additives leads to a highly localized substrate concentration and favorable conditions for the growth of microbes and rapid gas production in a reactor, such as a suitable pH, and the inhibition/promotion of acetogenesis and methanogenesis, etc.

3.1. Greenery biomass

Greenery biomass includes different plant extracts, plants, weeds, crop residues, and ensiled materials that are available naturally in the surroundings and are used as additives to improve biogas plant performance.

A series of plant extract materials containing naturally occurring steroids act as metabolic stimulants for microorganisms. Two such microbial stimulants, Aquasan and Teresan, have been used in previous studies. The addition of Aquasan and Teresan has been shown to be essential to improving anaerobic biodegradation and results in increased gas yields [44]. Powdered leaves of some plants and legumes (such as Gulmohar, *Leucacena leucocephala*, *Acacia auriculiformis*, *Dalbergia sisoo* and *Eucalyptus tereticornis*) have been found to stimulate biogas production by 18% to 40% [45,46]. The contribution of alkali-treated (1% NaOH for 7 days) plant residues (lantana, wheat straw, apple leaf litter and peach leaf litter) as a supplement to cattle dung resulted in methane concentrations of 63.6%, 58%, 59.6%, and 57.7% in the produced biogas, respectively, relative to that of cattle dung at 56.1% [47]. Crop residues such as maize stalks, rice straw, cotton stalks, wheat straw and water hyacinth, each enriched with partially digested cattle dung, enhanced gas production in the range of 10–80% [45,46]. The addition of *Parthenium hysterophorus* and soya sludge to cattle dung digesters has been observed to improve the methane content, gas production, manurial value and capillary suction time [48]. Increases in biogas production of 13.38%, 25.27%, 39.16%, 52.26% and 63.44% were observed with the addition of 10%, 15%, 20%, 25% and 30% mustard meal/cake addition in cattle dung digesters [49]. Sisal fiber waste showed COD removal efficiencies in the range of 80–93% at OLRs in the range of 2.4–25 g COD/L/d [50]. Furthermore, ensiled material demonstrated higher methane contents than fresh matter.

3.2. Biological additives

3.2.1. Fungi

Fungi, particularly those that attack ligning, are mainly used in the pretreatment of lignocellulosic biomass for biogas production. Several fungi classes, including brown-rot, white-rot and soft-rot fungi (i.e., *Ceriporiopsis subvermispora*, *Auricularia auricula-judae*, *Trichoderma reesei*), and basidiomycete fungi (e.g., *Ischnoderma resinosum* and *Fomitella fraxinea*) have been used for pretreatment with white-rot fungi being the most effective through the action of lignin-degrading enzymes (e.g., peroxidases and laccase) [51,52]. After fungal pretreatment, a 5 to 15% increase in the methane yield was obtained [53–55].

3.2.2. Microbial consortium

In contrast to fungal activity, a microbial consortium mainly increases cellulose and hemicellulose availability and thus digestibility. The consortium contains yeast and cellulolytic bacteria, heat-treated sludge, *Clostridium thermocellum*, and a mixture of fungi and composting microbes. Previous studies have reported methane yield improvements of 25–96.63% by using microbial consortia [56,57]. Although the addition of homo- and heterofermentative strains has shown positive effects on biogas yields, the combination of these strains with enzymes or bacteria or yeasts has shown even better performance [58]. *Phanerochaete chrysosporium* and cellulolytic strains of bacteria such as actinomycetes and mixed consortia have been observed to enhance gas production [45,46].

However, no matter fungi or microbial consortium, the challenge for a microbial agent used as an additive during AD is the strict requirements to the composition, the activity and the purity of strains and the sealing of reactors. Therefore, the investment cost of using this type of accelerator is high which would prevent its popularization and application.

3.2.3. Enzymes

Enzymes obtained from different microorganisms and plants are critical for substrate degradation by bacteria due to biochemical catalytic reactions. As a microbial supplement, enzymes can ensure the optimal growth and activity of various types of microorganisms, and therefore, biomass can be more resistant to shock loading. Enzymes have also been used to overcome drawbacks associated with the use of conventional chemical catalysts. The most commonly used enzymes include cellulase and hemicellulase [52]. The activities of some exoenzymes, such as proteases, lipases, and chitinases, have been reported in the literature [59]. However, in most cases, the effect of enzymes on enhancing biogas production is in a lower range of only 0–34% increases in methane yield have been achieved [52]. In addition, the cost of enzymes is high; therefore, the application of enzymes in pretreatment has been limited.

3.3. Inorganic additives

3.3.1. Chemical reagents

Chemical reagents are predominantly used for pretreatment of lignocellulosic materials due to their low-cost and high efficacy. By changing the properties of raw material, e.g., increasing the surface area, removing or dissolving lignin and hemicellulose, and reducing the crystallinity of cellulose, chemical reagents make lignocellulosic biomass more biodegradable and accessible to anaerobic microorganisms.

3.3.1.1. Alkali reagents. Alkali treatment breaks the links between lignin monomers or between lignin and polysaccharides, which make the lignocelluloses swell through salivation and saponification reactions, thus stimulating the lignin solubilization, the removal of hemicellulose, the disruption of interlinking ester bonds, and the neutralization of structural carboxylic acids. As a result, the specific surface area is increased and substrates become easily accessible to anaerobic microbes contributing to biogas production. Sodium hydroxide, potassium hydroxide, calcium hydroxide, magnesium hydroxide, ammonia and ammonium sulfite are common alkali reagents, and sodium hydroxide has been found to be one of the most effective alkalis for improving biogas production [60]. An increase in sodium hydroxide dosage led to an increase in cellulose and hemicellulose degradation with increased methane production. Lignin degradation in leaves was significant in reactors with 3.5% and 5.0% NaOH [61]. A ranking of alkali efficacy ($\text{NaOH} > \text{KOH} > \text{Mg}(\text{OH})_2$ and $\text{Ca}(\text{OH})_2$) was also observed [62]. In addition, another important aspect of alkali treatment is that some of the alkali can be consumed by the biomass itself, thus, higher concentrations of alkali reagents might be required to obtain the desired AD enhancement effect. Furthermore, alkalis help prevent decreases in pH during the acidogenesis process, increasing the efficiency of methanogenesis, and alkaline pretreatment performed at low moisture levels and temperatures is also particularly attractive.

3.3.1.2. Acid reagents. Acid reagents, including sulfuric acid, hydrochloric acid, phosphoric acid, maleic acid, formic acid and acetic acid, are highly desirable for lignocellulosic substrates. On the one hand, acid pretreatment results in the disruption of covalent bonds, hydrogen bonds, and Van der Waals forces, which consequently causes the solubilization of hemicellulose as well as the reduction of cellulose and the hydrolysis of hemicellulose into respective monosaccharides. On the other hand, acidic conditions are ideal for hydrolytic microbes. However, the loss of fermentable sugar obtained from the redundant degradation of complex substrates, the high costs of acids and the need to neutralize the acidic conditions before the AD process make acidic treatment less

attractive. At the same time, strong acids may result in the production of inhibitory by-products such as furfural and hydroxyl methyl furfural (HMF). Hence, strong acidic pretreatment is generally avoided. The use of dilute acids coupled with thermal methods is one possible alternative. Dilute acid makes cellulose and hemicelluloses more accessible to bacteria by breaking the linkage between polysaccharides and lignin. Moreover, dilute-acid pretreatment has been successfully used in hydrolyzing hemicelluloses, modifying the structure of lignin and increasing the cellulosic surface area [60,63]. For example, dilute acid pretreatment has been observed to increase the methane potential of sunflower oil cakes by up to 50%.

3.3.1.3. Oxidative reagents. The presence of oxygen accelerates the reaction rate and production of free radicals added to feedstock prior to pretreatment. Hydrogen peroxide is often favored during the pretreatment of lignocellulose substrates, and a significant enhancement in reaction efficiency has been observed, which is attributed to hydrogen peroxide's strong oxidation ability. The lignin content decreased by 6.7% to 32.0% when hydrogen peroxide was used, whereas the lignin content remained constant in the acid reagent-treated and untreated samples [64]. It should be noted that although high oxygen concentrations can yield faster reaction rates, high operating costs using pure oxygen are also generated. Therefore, air is usually also used as an oxidizing agent with the addition of water, a process referred to as wet oxidation pretreatment [52]. On the other hand, ozone is also a potent oxidant during the pretreatment (ozonolysis) of lignocellulosic biomass via lignin degradation. However, the application of ozonolysis focus on waste activated sludge and wastewater to improve digestibility, whereas few studies focus on the pretreatment of lignocellulosic biomass for biogas production in AD. Other peroxides, including fenton, peroxymonosulfate and dimethyldioxirane could increase biogas production but are not commonly used.

3.3.1.4. Inorganic salts. Several inorganic salts are widely used in stimulating biogas production, especially iron salts. Biogas production and the CH_4 content in biogas for cow dung and poultry litter AD were improved by adding FeSO_4 [65]. Methanogenesis was observed to have been enhanced by 40% and 42% by adding 20 mM FeSO_4 to daily-fed cow dung and poultry litter waste digesters [65]. Addition of FeCl_3 during the AD process resulted in an increase of more than 60% of biogas production; FeCl_3 is also efficient for the removal of hemicelluloses. Furthermore, the addition of FeCl_2 during batch experiments with swine excreta was reported to counteract sulfide inhibition [66], and addition of 20 mM sulfate was observed to enhance biogas production two-fold [67].

Among these chemical accelerants, alkaline and oxidative reagents are more efficient than acids in altering the structure of lignin, solubilizing the hemicellulose fraction, increasing the accessible surface area for microbes by swelling and partial decrystallization of cellulose and preserving most of the carbohydrates, particularly cellulose. However, although the use of chemical reagents is easier to manage, recycling the chemicals used for the pretreatment will be difficult and expensive preventing environmental pollution. Furthermore, the sustainability of reagents varies between different substrates, and they are not suitable for easily biodegradable substrates containing high amounts of carbohydrates due to their accelerated degradation and subsequent accumulation of VFA, which leads to failure of the methanogenesis step. In the context of these limitations, the most economically favorable and effective treatments, among those mentioned above, have yet to be identified [64].

3.3.2. Macro-nutrients and trace elements

For the AD process of biogas production, both macronutrients and trace elements are stimulatory and are more economically

Table 3
Stimulatory nutrients added into AD process of different substrates.

	Element	Feedstock	Function	Stimulatory concentration	Stimulatory effect
Micronutrients	Iron	Municipal solid waste	CODH, Precipitates sulfides	1000–5000 ppm (DB)	Promote organic degradation
		Energy crops			
	Nickel		CODH, other hydrogenases	0.029–27 mg/L	Aid hydrogenotrophic metabolism
					Reduce ammonia and sulfide toxicity
					Stabilize VFA levels
	Selenium	Animal excreta	F430, Benzoyl-CoA	0–10 mg/kg TS	Improve process stability
		Crops residues			Enhance the growth of microorganisms
		Food waste			Promote process start-up
	Tungsten	Stillage-fed	FDH	0.658–100 mg/L	Eliminate foam
	Zinc	Waste water	FDH, CODH, other Hydrogenases	0.0327–2 mg/L	Enhance methane content
Macronutrients	Chromium			4–15 mg/L	Improve biogas production rate
	Molybdenum		FDH	0.044–100 mg/L	
	Cobalt		Corrinoids CODH	0.029–5 mg/L	
	Carbon		Energy and cell material		
	Nitrogen		Protein synthesis		
	Potassium		Cell wall permeability	< 400 mg/L	
	Phosphorus		Nucleic acid synthesis	465 mg/L	
	Sulfur		Numerous enzymes	3.05–6.18 g/kg TS	
	Magnesium				

Note: db, dry basis, CODH, the enzyme carbon monoxide dehydrogenase, FDH, the enzyme formate dehydrogenase.

and environmentally sound accelerants compared with chemical reagents, which often require significant energy inputs. This stimulatory effect is significant, especially, for example, in the anaerobic digestion of energy crops, animal excreta, crop residues and the organic fraction of municipal solid waste (OFMSW), which lack these elements.

Microorganisms need trace elements as building blocks for growth, as well as to support enzymatic activities, chemical reactions and co-precipitation during the anaerobic digestion process. Fe reacts with H_2S to form FeS ; therefore, the addition of iron can be used to release corrosion in compressors and the toxicity of H_2S in biogas. Fe has also been identified as the most effective material for stabilizing food waste AD. Previous studies have reported that both macronutrients and trace elements have significant effects as additives during biogas digestion (Table 3) [68–71]. Ni is also stimulatory in biogas production. In a cattle dung batch study, Ni addition stimulated both biogas production and the methane content of biogas. The addition of Ca and Mg salts as energy supplements can enhance CH_4 production and prevent foaming. W is important for the degradation of propionate and methanogens [72]. The addition of Se and Co is indispensable to food waste AD stability and for operating at high ammonia concentrations [73]. Supplementation of Ca, Fe, Ni, and Co could be an alternative for releasing VFA accumulation [74]. Recently, commercial mixtures of trace elements have also drawn considerable attention for use in biogas plants. The stimulatory effects of mixtures with additions of Co and Ni, and Fe, Co and Ni have been observed to be significant. Supplementation with a mixture (Co, Mo, Ni, Se, and W) was reported to increase methane production to the range of 45–65% for inoculums with low background concentrations of trace metals [73]. These significant interaction effects have been most apparent for Ni, Fe, and Co. The solubilities of Cd, Cr, Pb, B and Se were also recently studied [75,76]. The addition of trace elements depends on many factors, e.g., substrate composition, metals contents, degradation mechanisms, operational parameters and the active microbial community. Furthermore, because the bioavailability of trace elements can vary with the concentration of trace element, the limited availability of these elements affects both process stability and biogas production and can inhibit the microbiological process.

Macronutrients, such as nitrogen, phosphorus, potassium, and magnesium, are required for the activation or functioning of many microorganisms in biological processes. Macronutrient

requirements are mainly assessed based on bacterial composition and growth yields and biomass composition. The nutrient ratio is generally C:N:P:S=600:15:5:1 and [10] the optimum C:N:P ratio for methane yield enhancement has been reported to be 200:5:1 [77]. During biological processes, carbon is usually supplied by the substrate and used to fortify a microorganism's cell structure. Nitrogen is needed for protein biosynthesis. Sulfur is necessary as a constituent of important amino acids and as an essential nutrient for methanogenic bacterial growth. The phosphate content is crucial for providing the energy carriers ATP and NADP during metabolism. However, one should account for the dramatic effect on biogas digestion performance that can be obtained by adding both macro-nutrients and trace elements together. In addition, although supplementation of micro-nutrients and trace elements could be a simple way to achieve AD process stabilization and efficient biogas generation, the economic feasibility of trace elements should be dependent on their cost.

4. Biogas AD reactors

4.1. Conventional anaerobic reactors

4.1.1. Anaerobic sequencing batch reactor

An anaerobic sequencing batch reactor (ASBR) is a single-tank fill-and-draw unit that utilizes the same tank for treatment and fermentation. Thus, all of the treatment steps and processes occur in a single basin or tank in an ASBR. During wastewater treatment, an ASBR is considered to be a good option for low-flow applications and allows for wider variations in wastewater strength. Compared with many continuous systems, the ASBR shows better process control and higher process efficiency. In terms of the major differences between ASBR and activated sludge continuous-flow systems is that the former carries out the functions of equalization, aeration, and sedimentation in a temporal sequence rather than in a spatial sequence. In addition, the ASBR system can be designed based on the ability of treating a wide range of influent volumes, whereas, a continuous system requires a fixed influent flow rate. The main advantages of an ASBR system are operational simplicity, efficient quality control of the effluent, flexibility of use, low input process and mechanical requirements,

cost-effectiveness and high biogas yield [78]. However, its poor self-immobilization and a certain amount of biological gas in the sludge cause insufficient settle-ability. Moreover, channeling and clogging as well as a larger volume are also limitations. Although attempts have been made to enhance biomass retention [78], ASBR operation requires some type of agitation to improve the transfer of the substrate to the microorganisms in the granulated biomass for anaerobic degradation. Therefore, there are many scientific features requiring further study to improve the operational performance of the ASBR.

4.1.2. Continuous stirred tank reactor

The continuous stirred tank reactor (CSTR) is the earliest (the first generation) high-rate anaerobic reactor. It is known for its reliability and is widely used to treat wastewater containing high-levels of suspended solids during an AD process, especially for the treatment of high-strength liquid animal manure and organic industrial wastes. In a CSTR system, microorganisms are suspended in the digester through intermittent or continuous mixing. Complete mixing offers good substrate-sludge contact with slight mass transfer resistance but consumes considerable energy and is labor-intensive as well [79]. The operation of a conventional single CSTR is simple but less efficient in terms of effluent quality. Therefore, a two-phase system appears to be the more common type of system. With respect to wet continuous digesters, the two-stage CSTR system is popular due to the simplicity of the system in design and operation and the its low capital costs compared with the one-stage CSTR. However, the system's sensitivity to substrates with high easily degradable organic loads and the complicated operation leads to fewer alternatives for improving digestion performance for the two-phase system [80]. In addition, the drawbacks associated with the system's structure and operation mode make it impossible to retain a high microorganism concentration in the reactor. In other words, microbial populations get washed out of the reactor along with the effluent. Due to mixing and continuous stirring, rapid acidification occurs, resulting in large VFA production, which could lead to AD process inhibition. Recently, advances have focused on CSTR variants to improve reactor performance through reactor volume optimization. Using the CSTR and a gravity sedimentation tank in series or in combination with a membrane bioreactor (MBR) could lead to a higher concentration of microorganisms in the reactor thereby providing more efficient digestion. In a laboratory-scale test, a serial CSTR was used in manure digesters [81]. The study found that the serial CSTR could improve biomass conversion efficiency and biogas yield primarily from the second reactor. It was explained that the second reactor helped utilize VFA produced from overloading in the first reactor, which improved the effluent quality and conversion efficiency. The two-phase anaerobic system of a CSTR for acidogenesis and an up-flow anaerobic filter for methanogenesis were used under different operating conditions for treating dairy wastewater. However, high suspended solid concentrations in dairy effluents particularly affect the treatment performance of anaerobic CSTRs and filters [82]. Vertical CSTR configurations are the most commonly used configurations in 90% of newly erected wet digesters.

4.1.3. Anaerobic plug-flow reactor

The anaerobic plug flow reactor (APFR) is another conventional process providing low concentrations of VFA in the effluent, a high degree of sludge retention and stable reactor performance. Plug-flow reactors are long, linear troughs usually situated above ground. This type of reactor is attractive in terms of efficiency and overall bioconversion compared to the conventional single-phase CSTR. The APFR features no internal agitation and is loaded with thick

manure of 11–14% total solids and works well at mesophilic or thermophilic temperature. The retention time is usually 15–20 days [77]. When using treated semi-solid waste, this type of reactor was used to provide low initial investment cost, high efficiency and relatively simple operation and maintenance [83]. Therefore, in both industrialized and developing countries, the reactor has significant potential to produce biogas. Additionally, the reactor has been tested experimentally using substrates, such as pig manure, distillery waste water, cattle residues, organic fraction of garbage and urban organic waste, etc. In many studies, a CSTR+ plug flow configuration was used, but associated equipment is also needed including standard structures (biomass storage tanks, the homogenization and feeding system, digestion tank and mixing system, gas cleaning, cogeneration unit and digestive tank)+biogas desulfurization/hydrolytic pretreatment [84].

4.2. Sludge retention reactors

4.2.1. Anaerobic contact reactor

The anaerobic contact reactor (ACR) is mostly employed for effluents with high concentrations of suspended solids. In some cases, high-rate mesophilic ACRs have been demonstrated to be a sustainable technology for a wide range of industrial effluents, for example, those found in food industry wastewater [85] and pulp and paper mills [86]. These reactors present similar features to their aerobic counterparts, i.e., activated sludge systems. Due to the fluid pattern of the reactor, inefficient mixing conditions that may reduce treatment capacity caused by heterogeneity within the biomass can be avoided [86]. Two main components are an agitated reactor and a solids settling tank for recycling of microorganisms. In the latter, settled sludge is recycled back into the main reactor. The substrate retention time and the degree of contact between the influent substrate and microorganism communities are the primary parameters affecting the performance of the reactor. Previous studies have proven that this type of reactor design is a remarkably efficient anaerobic process for the decomposition and conversion of organic matter to biogas. In terms of mass transfer rate, ACR is more advantageous than conventional anaerobic reactors such as up-flow anaerobic sludge blanket (UASB) reactors. Reported advantages include the contact process, rapidly achieved steady-state times due to mixing, sufficiently short hydraulic retention times and relatively high effluent quality, less effected by shock loading, favorable pH and limited biomass washout and change in biogas concentration and composition [85,87]. Additionally, the system can tolerate OLR of up to 8 kg COD/m³/d and can obtain COD removal efficiencies of approximately 78–95% [85,87]. In this type of reactor, the content of the reactor is completely mixed and then separated into a clarifier, a vacuum flotation unit or a lamella clarifier, and the supernatant is discharged as effluent. Settled anaerobic sludge is then recycled to seed the incoming influent. The sludge produced in this type of reactor will have a flocculent, like and dilute nature and consequently a limited organic loading rate.

4.2.2. Up-flow anaerobic sludge bed reactor

The up-flow anaerobic sludge blanket (UASB) reactor is a common, simple, compact and inexpensive technology used extensively for the treatment of effluent. The main structure of the reactor is a dense sludge bed located in the bottom, which guarantees good wastewater-biomass contact. Among the notable advantages, UASB requires less reactor volume and space, features higher flow velocity and biogas production and accommodates significantly higher organic load rates compared to flocculent sludge bed reactors. Furthermore, effluent recycling is not necessary because sufficient contact between wastewater and sludge is

guaranteed even at low organic loads with the effluent distribution system. Moreover, the availability of granular or flocculent sludge allows handling of higher COD loading rates and providing adequate treatment at lower HRTs than is possible with the anaerobic filter (AF). The main problem of filter clogging by suspended bacterial growth associated with AFs is not a major issue. To retain the microbial consortium in the reactor, UASB reactors can use dense bacterial granules, which serve as a filter to prevent bacterial washout and which also provide a larger surface area for faster biofilm development and improved methanogenesis. However, what is most noteworthy is that, due to the high performance dependent on suspended growth, the UASB's performance is largely dependent on the granule quality of its sludge. Additionally, when changing the waste type, the sludge granules will likely not retain their characteristics excepting a given type of waste. As a consequence, the challenge for this technology is that certain wastes result in a granular sludge quite readily, whereas other wastes behave slowly and some not at all. Moreover, a long start-up period and significant wash-out of sludge during the initial phase of the process are typically necessary, and the reactor requires skilled operation. Recently, modified reactor configurations have been proposed and successfully implemented to overcome these different constraints [88,89]. When a two-stage (acidogenic sequencing batch reactor+methanogenic UASB reactor) fermentation process was used to treat starfish, a conversion of 44% of the organic content of whole starfish to CH_4 was achieved [90].

4.2.3. Up-flow anaerobic solid-state reactor

The up-flow anaerobic solid-state (UASS) reactor was primarily demonstrated in 2010 [91]. Previous experiments have demonstrated the successful application of this type of reactor. With a quadruple two-phase, two-stage reactor design, each of the four systems consists of an up-flow anaerobic solid-state reactor, and an AF to prevent VFA accumulation was used to continuously ferment lignocellulosic biomass. The results revealed that the operation of the system was technically feasible in a long-term process without significant disturbances. Using maize silage as feedstock, the UASS reactor showed the highest methanogenic performance for the digestion of solid biomass, and the reactor was able to operate at OLRs of up to 17 g VS/L/d with methane yields of 312 L/g VS. Despite certain disadvantages with respect to anaerobic digestion, it was observed that the UASS is feasible to ferment straw [92]. These promising results prove that this system exhibits higher processing efficiency and a higher volume loading rate, lower investment cost, and simple operation and management. However, the system is limited by its structure and the monomer volume is small. The operating principle of this reactor is based on the spontaneous solid-liquid separation caused by natural differences in the densities of the substrate and process liquid [93].

4.2.4. Anaerobic baffled reactor

The anaerobic baffled reactor (ABR) was developed in the early 1980s and was initially introduced by McCarty and coworkers at Stanford University [94]. It consists of a series of compartments in one reactor, which is baffled to force incoming wastewater up through a series of blanked sludge. Thus, by rearranging the baffles and flow patterns, a large variety of ABRs can be set up to meet different needs. Over the two decades following its introduction, ABR had been found very limited popularity, but in recent years, its advantages are coming to the fore [90]. Operating with granules or internal media can enhance the system's stability. Therefore, the SRT can be separated from the HRT, achieving good COD and solids removal, low sludge production, and a small footprint. Bacteria

within the reactor gently rise and settle due to the flow characteristics and gas production in each compartment but move horizontally down the reactor at a relatively slow rate, giving rise to a cell retention time (CRT) of 100 days at an HRT of 20 h [90]. The most significant advantage of the ABR is its ability to separate acidogenesis and methanogenesis longitudinally down the reactor; different bacterial groups are thus allowed to develop under the most favorable conditions. In addition, the ABR has shown the potential for providing high efficiency at high loading rates and to be suitable for extreme environmental conditions and inhibitory compounds. Other advantages include higher tolerance to hydraulic and organic shock loads, longer biomass retention times and lower sludge yields compared to those of high-rate anaerobic reactors [95]. The ABR also overcomes the risk of clogging and sludge bed expansion, which plague other systems such as the AF and the conventional UASB reactor. Drawbacks include microbe wash-out from digesters, inadequate mixing and settle-ability of the microbial granules within the reactor, and incompatibility with certain types of wastewater.

4.2.5. Internal circulation reactor

The internal circulation (IC) reactor is in effect two UASB reactors working in tandem. Hence, an IC reactor contains two sets of 3-phase separation modules, whereas a USBR or EBR reactor only has one separation module. Due to this difference, the IC reactor can separate the gas, the liquid and the biomass simultaneously, improving biomass retention, which allows for higher biomass activity and improves the final effluent quality, and COD removal is also achievable. Additionally, due to the higher effluent quality, lower cost and higher efficiency can be achieved in the polishing step, and the extra mixing cost will be reduced. Separation of biogas at two different stages and internal effluent circulation are the special features of this reactor [90]. Based on its internal circulation, this new type of UASB reactor has a much higher OLR, with an OLR of up to 35 kg COD/m³/d being the highest achieved. Furthermore, compared to UASB and EGSB reactors, the IC system can treat low-strength wastewaters at higher HRT because of the up-flow liquid and gas velocities. This digester, under optimal hydrodynamic conditions, demonstrated good degradation capacity and buffer capability to resist various shock loadings. The IC reactor has been successfully applied in different industries, such as the brewery and beverage industry, pulp and paper industry, distillery and fermentation industry, and chemical and petrochemical industries [96].

4.3. Anaerobic membrane reactors

4.3.1. Anaerobic filter reactor

The anaerobic filter (AF) was initially developed to provide a support medium for the intimate contact between the influent and the bacterial mass, thus allowing for a biomass retention time longer than the HRT. On the medium, a biofilm is generated that supports the biomass separated from the effluent in this reactor configuration. The filters can be operated under either an up-flow or down-flow condition. The up-flow condition contains a high concentration of suspended biomass forming a biofilm in the structure of the fixed bed. The down-flow condition contains a high concentration of inorganic sulfur between the amount of biological oxygen demand (BOD) and low inorganic compound. Recycling can be applied for very high-strength wastewaters. Thus, the AF demonstrates excellent adaptability for biomass to a new carbon source and to organic load fluctuations and can also utilize dilute feeds. Compared to an anaerobic contact process, the AF demonstrated better performance and elimination of mechanical mixing and sludge settling and return for olive mill wastewater

fermentation. The successful application of the AF has been reported. The AF is a much simpler solution for industrial applications in particular. When treating the wastewaters discharged by raw milk quality control laboratories using the AF, COD removal higher than 90% could be obtained with, no biomass wash-out, and most of the fat contained in the wastewaters was successfully degraded [97]. Another study concluded that an AF packed with synthetic high surface area trickling filter media was a promising candidate for treatment of low-strength wastewaters and that post-treatment of sulfides and ammonia may be necessary. However, operational problems such as clogging of filter media limit the development of the system. To solve this restriction, a module of inclined tube settlers (ITS) was integrated with a fixed bed fixed film (FFFB) anaerobic reactor, i.e., an AF was used as a solid–liquid separation system while treating municipal wastewater. The results indicated that the suspended solids concentration was reduced by as much as 95% without any pretreatment, and a specific biogas yield of 0–0.35 m³ CH₄/kg COD_r with a 70% CH₄ content in the biogas was obtained [98]. Accordingly, the AF could be more suitable to treat wastewaters with lower suspended solids. The higher investment cost should also be considered in applications.

4.3.2. Anaerobic fluidized bed reactor

For an anaerobic fluidized bed reactor (AFBR), the medium for bacterial attachment and growth is small-inert particles, such as fine sand or alumina, kept in suspension by a rapid upward flow of incoming wastewater. This configuration allows for higher OLR and greater resistance to inhibitors. In addition, growth of a thin biofilm on these media particles and good attachment to biomass allow for good mass transfer efficiency in the AFBR. Furthermore, compared to AF technology, fluidized bed technology is more effective. The latter eliminates bed clogging, allows for hydraulic head loss combined with better hydraulic circulation and a greater surface area per unit of reactor volume, and the investment cost is much lower due to reduced reactor volumes. Moreover, the ability to remove suspended solid particles of domestic wastewater using the AFBR is better than that of the UASB. This type of reactor is more effective for the treatment of soluble, or suspended material feed that is easily biodegradable such as whey, whey permeate, black liquor condensate, etc. [99]. An AFBR was used to investigate and compare the treatability of very high suspended solids with different biodegradable particulate fractions and COD fractionation, thin stillage, as well as primary sludge derived from municipal wastewater treatment [100]. The results showed that maximum methane production yields of up to 0.31 L_{CH₄}/g COD and 0.25 L_{CH₄}/g COD were achieved for thin stillage and primary sludge, respectively. Based on the concept of the AFBR and the anaerobic membrane bioreactor (AnMBR), the applications of the anaerobic fluidized bed membrane reactor (AFBMR) have recently been considered for anaerobic process [101]. However, membrane fouling is a constraint for these reactors. It has been reported that proteins are the dominant contributors to membrane fouling at low temperature [102]. To eliminate or reduce membrane fouling, researchers have demonstrated that a fraction of solid media such as granular activated carbon (GAC) or powder activated carbon (PAC) can be added, because they can effectively adsorb microbial metabolic products [103] as well as fluidized GAC [101].

4.3.3. Expanded granular sludge blanket

The expanded granular sludge blanket (EGSB) is defined as a modification to the UASB reactor and is generally used when the volumetric gas production rate is low and mixing in a UASB reactor by up-flow velocity alone is insufficient. As a derivative of the UASB, the EGSB responds to the needs of small and medium-sized

industries in treating low-strength soluble and complex wastewaters. The EGSB reactor is distinguished by several advantages over the UASB: (i) The EGSB offers a smaller footprint, higher mixing due to the higher up-flow velocities and consequently improved mass transfer, biomass activity and better transport of substrate into sludge aggregates. (ii) The reactor features higher organic and hydraulic loadings, especially for acidified wastewater under psychrophilic conditions, even at temperatures as low as 10 °C [104]. (iii) The EGSB is capable of treating wastewaters containing lipids and toxic/inhibitory compounds. Other researchers have compared the two reactors with the EGSB performing better than the UASB. (iv) The EGSB is more suitable for soluble pollutant treatments, especially for low-strength wastewater. However, suspended solids cannot be substantially removed.

A comparison of the above-described reactors reveals that the EGSB and IC are the most advanced AD reactors as derivatives of the UASB and the most efficient, especially for medium concentration (COD < 1000 mg/L) wastewater. In terms of the prominent advantages of EGSB and IC, the systems show higher organic loading, higher resistance to impact, up-flow velocity and sufficient attachment between sludge and biomass. Moreover, they all contain 3-phase separation modules that can separate the gas, the liquid and the biomass simultaneously; therefore, devices for precipitation separation, auxiliary degassing and reflux are not required. Consequently, investment and operating cost savings can be realized.

5. Biogas AD processes

The standards for biogas AD process type factors include AD temperature, substrate TS concentration, AD phase and feeding methods. Herein, the processes are overviewed based on the type of substrates used, including lignocelluloses waste, municipal solid waste, food waste, livestock manure and waste activated sludge.

5.1. Lignocellulose waste

Lignocellulose wastes mainly include crop residues and logging residues with crop residues making up the majority. For China, more than 800 million metric tons of waste agricultural straw is produced per year [105]. However, this waste cannot be digested by itself due to recalcitrant materials (lignin, cellulose and hemicellulose) that result in low biodegradation and poor digestion performance; thus, extra accumulative measures are needed to start the digestive process such as pre-treatment and inoculums. For an inoculum, the enzyme activities lead to higher substrate degradation and biogas production. Another characteristic of an inoculum is that the nutrient contents can enhance the enzyme activity and biogas production [71]. However, lignocellulose wastes have a high C/N ratio, which leads to a decrease in biogas; thus, co-digestion of lignocellulose wastes and other organic matter is often reported. Previous studies concentrated on AD processes of crop straws and combined these processes with various pretreatments prior to the main AD process, digestion conditions largely involved co-digestion with other organic matter, mesophilic and single batch-digesters (Table 4) [64,106–115].

5.2. Municipal solid waste (MSW)

The rapid economic growth and expansion of urbanization and industrialization, the rise of mega-cities, and increasingly affluent lifestyles, coupled with accelerated product obsolescence and ubiquitous wastefulness tendency, all draw considerable attention toward municipal solid waste (MSW) management as the collection and disposal of MSW is also a major urban environmental

Table 4
AD process of the lignocellulose waste.

Feedstock	Pretreatment	Substrate	ADT ^c (°C)	DT ^d (d)	Stage	Reactor	BP ^e (mL/ g VS)	MP ^f (mL/ g VS)
Rice straw	Screened 8-mesh sieve [106]	Inoculum-to substrate-ratio of 0.5 ^a	37	40	Single	Batch digesters	325.3	178.3
	3–5 mm [107]	Initial C/N ratios of 20–30, adding 465 mg-P/L	22 ± 2	120	Single	Batch digesters	350	290
Corn straw	20–30 mm, 3% H ₂ O ₂ (w/w) [64]	8% TS content	37 ± 1	35	Single	Batch digesters		216.7
	5 mm, 5% NaOH (w/w) [108]	C/N ratios of 18, 22% TS content	37	40	Single	Solid-state AD reactor	372.4	
	3 cm [109]	C/N ratios of 20, Co-digestion with chicken manure (1.4:1 based on VS), 12% TS content, OLR of 4 g VS/L/d	37	140	Single	CSTR		223
Wheat straw	100 °C, 10% g NaOH/g TS [110]	C/N ratios of 18, 22% TS content	35 ± 0.5	31	Single	Batch digesters		Enhanced up to 67% 269
	10 mm [111]	Co-digestion with spent mushroom substrate (1:1 based on VS), 16% TS content	37	62	Single	Batch digesters		
	screened 16-mesh sieve 6% NaOH (w/w) [112]	5% TS content, substrate to inoculums ration of 4, Co-digestion with swine manure (50/50)	35	45	Single	Batch digesters	449	296
Fallen leaves	9 mm, 5% NaOH (w/w) [108]	Substrate to Inoculum ratio of 4:1, C/N ratios of 18, 20% TS content	37	30	Single	Batch digesters		82
Asparagus stem	0.28–0.45 mm, 6% NaOH (w/w) [114]	Substrate to Inoculum ratio of 1	35	60	Single	Batch digesters		242.3
Yard trimmings	12.7 mm and 60% moisture content, white-rot fungus <i>C. subvermispura</i> (ATCC 96608) [115]	Substrate to Inoculum ratio of 4:1, 20% TS content	37 ± 1	40	Single	Batch digesters		44.6

^b Mixed liquor suspended solids.

^a Based on the VS content.

^c Anaerobic digestion temperature.

^d Digestion time.

^e Biogas production.

^f Methane production.

Table 5
AD process of the municipal solid waste (MSW).

Feedstock	Substrate	ADT (°C)	AD (d)	Stage	Reactor	MP (mL/g VS)
OFMSW ^a	20% TS content, 8–5 days SRT [117]	35	200	Single	Semi-CSTR	340
	Co-digested with manure (1:5 and 1:1.43), OLR of 3.3–4.0 g VS/L/d, process liquid recirculated [118]	55	336	Single	Batch digester	630–711
	20% TS content, 30% of volume [119]	55	90	Single	Batch digester	50–180
	Co-digested with cow manure, C/N of 20:1, (63.7% paper, 18.2% food waste, 9.1% grass clippings, and 9% cow manure) [120]		150	Two		172 m ³ /t of dry waste
MSW	20% TS content, 8–5 days of SRT [117]	55	200	Single	Semi-CSTR	330–340
	MSW leachate seeding with granular sludge of paper mill [121]	35	50	Two	IIEC ^b	11.77
	Lignocellulose from municipal solid waste (1:1:1 ratio of office paper, newspaper, and cardboard), 5.0% substrate concentration pretreated by microbial consortium (MC1) for 6 days, 1:1 ratio of substrate to inoculums [122]	35		Single	Batch digester	41
	Leachate was recirculated and added Na ₂ CO ₃ , NaHCO ₃ and NaOH [123]	25	90	Single	Simulated landfill reactors	

^a Organic fraction of municipal solid waste.

^b Integrated internal and external circulation (IIEC) reactor.

issue in the world today. Recently, MSW has been widely used to generate waste-to-energy by conventional technology. In Bangladesh, using a landfill gas recovery process, the generation of electricity from MSW in six mega-cities is ~186,408 kW h/d [116]. Currently, there are many technologies available to treat MSW usually conducted in the range of 20% to 40% TS. The AD process is playing a vital role for MSW (Table 5) [117–123] process typically known as dry AD. The main advantages of this process are less reactor volumes and lower consumption of water and energy. However, the slower the AD process is, the more robust equipments must become and the higher the concentration of toxic compounds generated is, which represent two significant drawbacks.

5.3. Food waste

Food waste (FW) is mainly produced by hotels, restaurants, families, canteens and companies. With population and economic growth, this type of waste has rapidly increased and accounts for a large part of MSW. The amount of FW was approximately 90 million tons in China in 2010 [124]. The TS and VS contents of FW are in the ranges of 18.1–30.9 and 17.1–26.35, respectively [124]. Therefore, the moisture content is high, and consequently, FW can be easily employed for biodegradation. Compared with traditional approaches such as landfill disposal, incineration and aerobic composting for FW treatment and valorization, AD conversion of FW to biogas is an effective solution because the organic matter in

Table 6

AD process of the food waste (FW).

Feedstock	Substrate	ADT (°C)	DT (d)	Stage	Reactor	BP (mL/g VS)	MP (mL/g VS)
Fruit and vegetable wastes	3 kg/d of daily loading rate, 27 days of hydraulic residence time, 2.5 to 3.0 kg VS/m ³ /d of OLR [127]	35 ± 0.5	180	Single	Pilot-scale anaerobic digester	780	430
Food and green waste	Food and green waste and their mixture 1:1 based on the VS and substrate to inoculum ratios of 4.0, 1.6, 1.6, respectively. [128]	52 ± 2	25	Single	Batch digester	784,631,716	518,357,430
FW and dairy manure	3 g/L of initial VS, 52/48% mixture ratio of unscreened manure and FW, 20 days of HRT [126]	35	30	Single	Batch digester		311
FW	Substrate to inoculum ratios of 1:1, 12 days of HRT [129]	55	30	Three	Batch digester		223 (sCODdegraded)
FW	FW consisted of rice (4.0% w/w), noodles (2.5%), bread (1.7%), tea leaves (8.0%), vegetables (53.6%), fruit (24.8%), meat (2.2%), fish (2.7%), and egg shells (0.5%), OLR of 9.2 kg VS (15.0 kg COD) m ⁻³ /d [130]	37 ± 1	225	Single	Semi-continuous digester	6.6 L/L/d	455
FW and dairy manure	A mixture of 70% manure, 20% food waste and 10% sewage sludge, 4% of TS content, OLR of 1.2 g VS/L day [131]	36	160	Single	CSTR		603
FW and cattle manure	FW to cattle manure ratio of 2 [125]	35 ± 1	30	Single	Batch semi-continuous digester		388,317
FW and yard waste	An feedstock/effluent ratio of 1 with 20% FW and an feedstock/effluent ratio of 2 with 10% FW [132]	36 ± 1	30	Single	Batch digester		67% and 65% of methane contents
Fruit and vegetable waste	Step-wise increase in the loading rate from 5 kg VS/m ³ /d to 10 kg VS/m ³ /d [133]	35	113	Two	Batch digester		330 mL/g COD _{fed}

Table 7

AD process of the livestock manure.

Feedstock	Substrate	ADT (°C)	AD (d)	Stage	Reactor	BP (mL/g VS)	MP (mL/g VS)
DM^a	Co-digested with straw residues of mixing ratios at 1:9, 3:7, 5:5, 7:3 and 9:1, 8.0% of TS content, with a OLR of 3.2 g/L every two days [134]	35	40	Single	Semi-continuous digester	220–525 mL/d	
	Co-digested with SM ^b , photo-dark, 8% of TS content [135]	35 ± 2	35	Single	Batch digester	15,447.5 mL (cumulative)	
	15 Days of HRT, 1 atm of operating pressure [80]	55	190	Single	Semi-serial CSTR.	11% higher than a single CSTR	
	Filtrate through a1-mm sieve, 22.5 days of HRT [136]	55	12	Single	UASB		300 (mL/g COD)
	Co-digested with fruit and vegetable wastes 50:50 (wet weight) at 5.01 kg VS/m ³ /d, 21 days of HRT [137]	35	56	Single	CSTR		450
	SM with mechanical (separation of liquid and solid matrix by using a 0.25 mm pore size screen), chemical (flocculant agent and strong chemicals), and thermal (170 °C for 30 min) pretreatments, a ratio substrate COD/inoculum VS of 0.6 [138]	32	30–60	Single	Batch digester		212–412 (mL/g COD)
	SM with ammonia concentration of 6 g-N/L, 15 days of HRT	37		Single	CSTR		188
	Addition of 1.5% (w/w) activated carbon, 10% (w/w) glauconite or 1.5% (w/w) activated carbon and 10% (w/w) glauconite [66]	55	60	Single	Batch digester		An increase to 126,90, 195
GM^c	80% SM co-digested with glycerine [66]	35	30	Single	Batch digester		215 (mL/g COD)
	Co-digested with corn stalks and rice straw (30:70 and 70:30), 8% of TS content [27]	35 ± 1	55	Single	Batch digester	14,840–16,023 mL (cumulative)	
PM^d	Mixture of llama–cow–sheep manure, the maximum OLR value of 4 to 6 kg VS/m ³ /d [139]	18,25	100	Single	Semi-continuous batch digester		47 to 55%
	Removed ammonia CM ^e used a rotary evaporator with an ammonia-stripping unit 55 °C and pH 8, biogas recycle [140]	55 ± 2	20	Single	Batch digester		195
	Semi-solid (10% TS) ammonia stripped CM co-digested with agricultural wastes (7:3), substrate to inoculum ratio of 1:3(V/V) [141]	55 ± 2	165	Single	Semi-continuous batch digester		695
	A 10% fraction of PM co-digested with SM and sewage sludge (10:20:70) (w/w) at feed VS of 42.95 g/kg and SRT of 15 days [142]	35 ± 1	90	Single	Semi-continuous batch digester	336	

^a Dairy manure.^b Swine manure.^c Goat manure.^d Poultry manure.^e Chicken manure.

Table 8
AD process of the waste activated sludge (WAS).

Feedstock	Pretreatment	Substrate	ADT (°C)	AD (d)	Stage	Reactor	BP (mL/g VS)	MP (mL/g VS)
WAS	Microaerobic, 60–70 °C, 1 d [144]	WAS	37	10	Single	Batch digester	Increase 50%	
	Alkaline-method (pH12) and was stirred at 80 rpm for 6 h [143]	The mixture of alkaline-pretreated sludge and seed sludge with a ratio of 9:1, adding the zero valent iron of 20 g/L	35 ± 1	20	Two	Batch digester		Increase 43.5%
	Oxidation pretreatment (0.15 g O ₃ /g TS) [145]		35	18	Single	Batch digester	Increase 145%	
	High pressure pretreatment [146]	Pretreated sludge, 15 days of SRT	35 ± 2	30	Single	Batch digester		
	ultrasonic pretreatment (200 W, 30 min, 20–25 °C) [147]	Ratio of the seed sludge to WAS 3:1	36 ± 1	12	Single	Batch digester		Increased by 64%,
	Thermal pretreatment at 175 °C, 30 min [148]	15 Days of HRT	35		Single	CSTR		Increase of 62%
	Thermal pretreatment at 175 °C, 40 min [149]	2.9 Days of HRT	37	120	Two	Fixed film reactor		

FW is well suited for anaerobic microbial growth. During the FW AD process, temperature, VFA and pH, C/N ratio, ammonia, long chain fatty acids and metal elements are the key parameters. Because most of the carbohydrate polymers and proteins in FW exist in solid form, pretreatment is also needed before AD. Although FW has a high potential for methane production, inhibition in single digestion always occurs because the nutrients are always imbalanced in the anaerobic digester. The inhibiting factors include insufficient trace elements and excessive macro nutrients [125,126], unsuitable C/N ratios and high lipid concentrations. To overcome these disadvantages, co-digestion of FW with other organic substrates has been demonstrated to be a promising approach. Various techniques for efficiently processing FW by AD have been presented in previous studies (Table 6) [127–133].

5.4. Livestock manure

Today, in most countries, intensive livestock farming is continuously developing. The abundance of livestock manure exceeds its demand as fertilizer and results in adverse impacts on both the environment and humans. However, the single AD of manure results in low performance due to nutrient imbalance and ammonia inhibition. Generally, livestock manure contains a high nitrogen content: fresh goat manure (1.01%), chicken manure (1.03%), dairy manure (0.35%) and swine manure (0.24%) [27]. Livestock manure can be used to balance the C/N ratio of single straw residues, and to obtain a suitable pH during AD with the production of ammonia. Hence, previous studies have focused on various AD processes (Table 7) [27,66,80,134–142], namely those for the co-digestion of livestock manure with other organic residues, which prevent the adverse impacts of livestock manure. Other co-digestion benefits include increased loading of readily biodegradable organic matter, dilution of toxic substances, improved buffer capacity of the mixture, higher biogas yield, better digested product quality, and reduced costs [142].

5.5. Waste activated sludge

In addition to the waste organic matter mentioned above, waste activated sludge (WAS) is also another type of resource for AD processing. With increasing industrialization, the amount of WAS is rising rapidly. Protein and cellulose are the two main components of this type of sludge, which can be biodegraded to produce biogas. Generally, the AD process of WAS includes three stages: (i) hydrolysis of biological polymers, (ii) conversion of hydrolysate to H₂ and

acetate and (iii) conversion of acetate and H₂ to methane [143]. Like other AD processes, hydrolysis is the rate-limiting step in the AD of WAS but can be overcome by improved rates. In addition, WAS shows relatively low degradability, especially at long sludge ages. Hence, to accelerate sludge digestion, various pretreatments have recently been implemented, including biological (largely thermal phased anaerobic), thermal hydrolysis, mechanical (such as ultrasound, high pressure and lysis), chemical with oxidation (mainly ozonation), alkali treatments, and co-treatment processes (Table 8) [143–149]. Biological pretreatment aims at intensifying the hydrolysis process before the main digestion process. Thermal hydrolysis application achieves partial solubilization of sludge by improving de-waterability [148]. According to previous studies, an optimal temperature range is 160–180 °C for processing times of 30–60 min [150]. However, other research has indicated that treatments at higher temperatures (170–190 °C) result in low sludge biodegradability despite high solubilization efficiencies. Mechanical treatment methods focus on providing moderate performance improvements with moderate electrical input.

6. Conclusions and recommendations

Today, the theory and technology of biogas production are mature and well developed: the key to further research is optimization. The appropriate factors affecting efficiency can create the basic living conditions for microorganisms due to their sensitivity to the environment. Although thermophilic AD has a rate-advantage over mesophilic digestion, a larger investment cost is needed to deploy thermophilic systems. Single-substrate AD results in a nutrient imbalance; thus, optimization of the C/N ratio for co-digestion is possibly the most cost-effective technique for reducing the relevant toxicity and is also the easiest to implement. To obtain the optimal OLR, decreasing the HRT is a possibility. However, published information on the maximum feasible loading rate is still lacking. Furthermore, by adding accelerants in the AD process, the digestion performance is greatly enhanced due to the adsorption of the substrate onto the surface of the additives. Greenery biomass is available naturally in the environment and generates no secondary pollution, thus this type of biomass is regarded as a promising accelerant. Nevertheless, greenery biomass has low efficiency. Biologically pure cultures have obvious benefits, but their cost is high and precise operational technologies are required. Inorganic additives overcome the disadvantages presented by both of these accelerants and offer high efficiency but can

lead to secondary pollution. Moreover, in comparing various reactors, EGSB and IC are observed to be the most advanced AD reactors as derivatives of the UASB and also the most efficient, especially for medium concentration (COD < 1000 mg/L) wastewater. In terms of the prominent advantages of EGSB and IC, the systems show higher organic loading, higher resistance to impact, up-flow velocity and sufficient attachment between sludge and biomass. Moreover, the systems contain 3-phase separation modules that can separate the gas, the liquid and biomass simultaneously; therefore, devices for precipitation separation, auxiliary degassing and reflux are not required. Consequently, investment and operational cost savings can be realized. With respect to the sustainability of biogas technology, developing optimal cost-optimal input/output ratio of digestion process could be a promising technology.

For the further development of biogas AD, the authors recommend that attention be given to the combination of two or more of the above mentioned factors affecting efficiency and accelerants to promote digestion process performance, especially for accelerants, which play a vital role in the AD process, in particular, studies should consider strengthening microbial metabolism and stimulating degradation of organic matter. Multi-stage systems, based on the feasibility of a single stage, may be used to achieve sufficient utilization of substrates, e.g., H_2 production followed by CH_4 production. H_2 production prior to the CH_4 production stage not only produces energy, but can also serve to pretreat the substrate without reducing the amount of biomass, effectively, advancing the startup of CH_4 digestion. Additionally, due to higher investment and operational costs, the development of biogas plants in household scale would be impeded. Consequently, the authors also recommend that the practicality of theoretical and technological studies based on laboratory scale-experiments be emphasized. This is because there is, to a great extent, a close relationship between the modernization and industrialization of agriculture and intensification of biogas development, which has led to changes in scale from household to medium- and large-scale biogas plants. Accordingly, better stability and operability are recommended for AD progress.

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